Premises or conclusions can be implicit in an argument.

- *A fallacious argument is an argument where the premises do not entail the conclusion.*

Text: "{complete_example_input}"
Based on the above text, determine whether the following sentence is part of a fallacious argument or not. If it is, indicate the type(s) of fallacy without providing explanations. The potential types of fallacy include:

- *appeal to positive emotion*
- *appeal to anger*
- *...*
- *guilt by association*
- *tu quoque*

Sentence: "{sentence_input}"
Output:

An example and the generated output using GPT-3.5:
Definitions:

- *An argument consists of an assertion called the conclusion and one or more assertions called premises, where the premises are intended to establish the truth of the conclusion. Premises or conclusions can be implicit in an argument.*
- *A fallacious argument is an argument where the premises do not entail the conclusion.*

Text: "I lost my phone in the living room, so it will always be in the living room when it is lost."
Based on the above text, determine whether the following sentence is part of a fallacious argument or not. If it is, indicate the type(s) of fallacy without providing explanations. The potential types of fallacy include:

- *appeal to positive emotion*
- *appeal to anger*
- *...*
- *guilt by association*
- *tu quoque*

Sentence: "I lost my phone in the living room, so it will always be in the living room when it is lost."
Output: This sentence is an example of the fallacy of hasty generalization.

I Metrics

Figure 5 and Figure 6 show an example for the calculation of our precision and recall metrics are computed. We now prove some properties of our metrics.

Proposition I.1. *Given a gold standard G, where each span comprises only a single sentence, and where each fallacy set contains only one element, and given a prediction P, where each span comprises only a single sentence, our precision coincides with the standard precision.*

Proof: By definition, we have, for any spans p, g, and for any sets l_p, l_g:

$$C(p, l_p, g, l_g, |p|) \tag{1}$$
$$= \frac{|p \cap g|}{|p|} \times \delta(l_p, l_g) \tag{2}$$
$$= \frac{|p \cap g|}{|p|} \times [l_p = l_g] \tag{3}$$

If p and g are singleton spans, this boils down to

$$= [p = g] \times [l_p = l_g] \tag{4}$$
$$= [p = g \wedge l_p = l_g] \tag{5}$$

Thus, we have, for any singleton span s and any label l:

$$[(s, \{l\}) \in G] \tag{6}$$
$$= [\exists (s', \{l'\}) \in G : s' = s \wedge l' = l] \tag{7}$$
$$= [\exists (s', \{l'\}) \in G : C(s, l, s', l', |s|) = 1] \tag{8}$$
$$= \max_{(s', \{l'\}) \in G} C(s, l, s', l', |s|) \tag{9}$$

This entails that the number of true positives (TP) is

$$|\{(s, l) \in P \mid (s, \{l\}) \in G\}| \tag{10}$$
$$= \sum_{(s,l) \in P} [(s, \{l\}) \in G] \tag{11}$$
$$= \sum_{(s,l) \in P} \max_{(s', \{l'\}) \in G} C(s, l, s', l', |s|) \tag{12}$$

The standard precision is the ratio of true positives (TP) out of the sum of true positives and false positives (FP):

$$\text{Standard Precision} \tag{13}$$
$$= \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

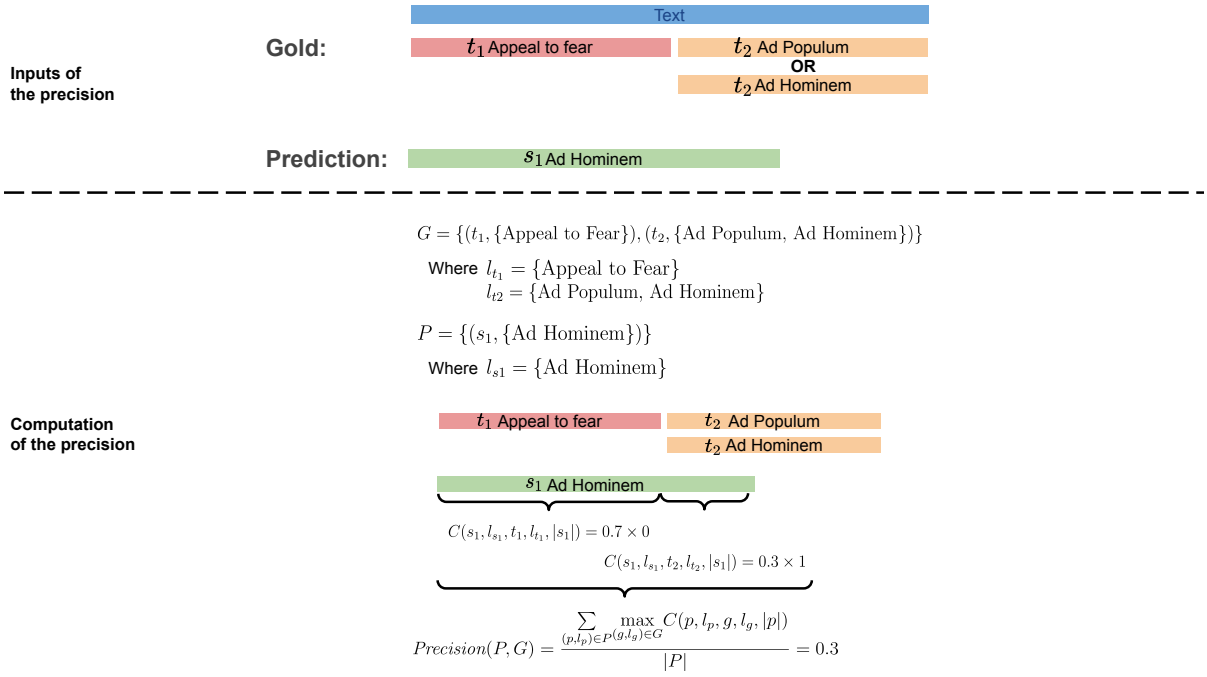

Figure 5: Example of Precision computation with alternatives.

With $|P| = \text{TP} + \text{FP}$ and Equation 12, this is equivalent to

$$= \frac{\sum_{(s,l)\in P} \max_{(s',\{l'\})\in G} C(s, l, s', l', |s|)}{|P|}$$

□

Proposition I.2. *Given a gold standard G, where each span comprises only a single sentence, and where each fallacy set contains only one element, and given a prediction P, where each span comprises only a single sentence, our recall coincides with the standard recall.*

Proof: As previously, for any p, g that are singleton spans, we have:

$$C(p, l_p, g, l_g, |g|) \quad (15)$$
$$= [p = g \wedge l_p = l_g] \quad (16)$$

Thus, we have, for any singleton span s and any label l:

$$[(s', l') \in P] \quad (17)$$
$$= [\exists (s, l) \in P : s = s' \wedge l = l'] \quad (18)$$
$$= [\exists (s, l) \in P : C(s, l, s', l', |s'|) = 1] \quad (19)$$
$$= \max_{(s,l)\in P} C(s, l, s', l', |s'|) \quad (20)$$

This entails that the number of true positives (TP) is

$$|\{(s', \{l'\}) \in G^- \mid (s', l') \in P\}| \quad (21)$$
$$= \sum_{(s',\{l'\})\in G^-} [(s', l') \in P] \quad (22)$$
$$= \sum_{(s',\{l'\})\in G^-} \max_{(s,l)\in P} C(s, l, s', l', |s'|) \quad (23)$$

The standard recall is the ratio of true positives (TP) out of the sum of true positives and false negatives (FN):

$$\text{Standard Recall} \quad (24)$$
$$= \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (25)$$

With $|G^-| = \text{TP} + \text{FN}$ and Equation 23, this is equivalent to

$$= \frac{\sum_{(s,\{l\})\in G^-} \max_{(s',l')\in P} C(s, l, s', l', |s|)}{|G^-|}$$

□

Proposition I.3. *In cases where a system predicts multiple labels for a single span, and the corresponding gold standard also contains multiple alternative labels for that span, the system's recall does not increase with the number of correctly predicted labels. Our recall formula ensures that multiple predictions for the same span do not artificially inflate the recall metric.*

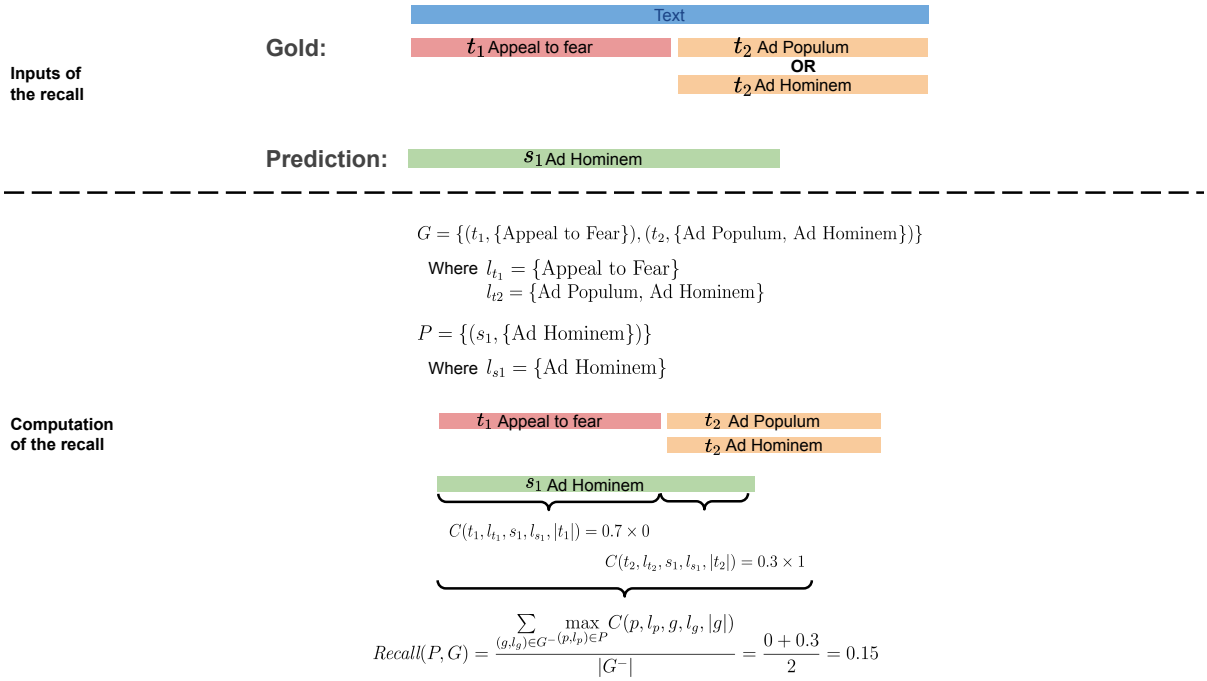

Figure 6: Example of Recall computation with alternatives.

Proof: By our definition, we have, for any prediction P and any gold standard G:

$$Recall(P,G) = \frac{\sum_{(g,l_g)\in G^-} \max_{(p,l_p)\in P} C(p, l_p, g, l_g, |g|)}{|G^-|} \tag{26}$$

By definition, we have, for any spans p, g, and for any sets l_p, l_g:

$$C(p, l_p, g, l_g, |g|) \tag{27}$$

$$= \frac{|p \cap g|}{|g|} \times \delta(l_p, l_g) \tag{28}$$

If p and g are the same spans, this boils down to

$$= \delta(l_p, l_g) \tag{29}$$

$$= [l_p \in l_g \neq \emptyset] \tag{30}$$

The max operation ensures that the score contribution for the span in the recall is based on the best match between predicted labels and gold standard labels, capped at 1 regardless of the number of labels correctly predicted. So even if multiple labels in l_p for p in P match with different alternative labels in l_g in G, the contribution to the recall for the span remains 1.

$$\max(C(p1, l_{p1}, g, l_g, |g|), \tag{31}$$

$$C(p2, l_{p2}, g, l_g, |g|), \tag{32}$$

$$, ..., \tag{33}$$

$$C(pn, l_{pn}, g, l_g, |g|)) \tag{34}$$

$$= 1 \tag{35}$$

□

I.1 Metric equivalence

The metric proposed in this paper is similar to the metric proposed in (Martino et al., 2019). This metric supposes that there is no overlap of spans with the same label. However, such spans are very frequent in a multi-level taxonomy, when evaluating Levels 0 and 1. Consider the following example:

Example I.1

You are a liar. Therefore you are wrong.

In this example, there is only one *abusive ad hominem*, which is a *Fallacy of Credibility* on Level 1. Now assume that the model outputs: (You are a liar, *abusive ad hominem*), (You are a liar, therefore you are wrong, *tu quoque*). Using the recall from (Martino et al., 2019), and $G = \{([0, 40], \text{CREDIBILITY})\}$, $P = \{([0, 10], \text{CREDIBILITY}), ([0, 40], \text{CREDIBILITY})\}$